

4.3.1 Lifestyle and health

The way in which people live their lives can have long-term consequences for their health. The chances that someone will be affected by conditions such as cardiovascular disease, diabetes or cancer may depend on lifestyle factors, including exercise, diet, alcohol consumption and smoking.

Treatments are available to control the symptoms of non-communicable diseases (see also [Preventing, treating and curing diseases](#) (page 60)) but the benefits have to be weighed against the risks.

The scientific understanding of the reproductive hormones can help people to control their fertility and also to receive treatment for infertility.

4.3.1.1 Health and disease

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the relationship between health and disease. Describe different types of diseases (including communicable and non-communicable diseases).	Health can be defined as 'a state of physical, mental and social well-being' and not merely the absence of disease. Factors including diet, stress and life situations can affect both physical and mental health. Diseases stop part of the body from working properly. This causes symptoms, which are experienced by the person affected by the disease. Communicable (infectious) diseases are caused by microorganisms called pathogens. They may infect plants as well as animals and are spread by direct contact, by water or by air. Non-communicable diseases, such as heart disease, cancer and diabetes, are the leading cause of death in the world.	

4.3.1.2 Risk factors for non-communicable diseases

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that many non-communicable human diseases are caused by the interaction of a number of factors. To include cardiovascular diseases, many forms of cancer, some lung and liver diseases and diseases influenced by nutrition, including Type 2 diabetes. Explain the effect of lifestyle factors, including exercise, diet, alcohol and smoking, on the incidence of non-communicable diseases at local, national and global levels.	Risk factors are aspects of a person's lifestyle, or substances present in a person's body or environment, that have been shown to be linked to an increased rate of a disease. For some a causal mechanism has been proven. Examples are: • the effects of diet, smoking and exercise on cardiovascular disease • obesity as a risk factor for Type 2 diabetes • the effect of alcohol on liver and brain function • the effect of smoking on lung disease and lung cancer • the effects of smoking and alcohol on unborn babies • carcinogens and ionising radiation as risk factors in cancer.	WS 1.5 Interpret data about risk factors, or about differences in the incidence of non-communicable diseases in different parts of the world. WS 1.4 Discuss the human and financial cost of these non-communicable diseases to an individual, a local community, a nation or globally. MS 4a Translate information between graphical and numerical forms. MS 2c, 4a Extract and interpret information from charts, graphs and tables. MS 2d Understand the principles of sampling as applied to scientific data in terms of risk factors. MS 2c Construct and interpret frequency tables and diagrams, bar charts and histograms. MS 2g Use a scatter diagram to identify a correlation between two variables.

4.3.1.3 Treatments for cardiovascular disease

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Evaluate some different treatments for cardiovascular disease.	In coronary heart disease layers of fatty material build up inside the coronary arteries. This reduces the flow of blood through the coronary arteries. This can lead to a heart attack. Statins are widely used to reduce blood cholesterol levels, which slows down the rate of fatty material deposit. Stents are used to keep the coronary arteries open. In some people heart valves may become faulty, preventing the valve from opening fully, or the heart valve might develop a leak. Faulty heart valves can be replaced using biological or mechanical valves. In the case of heart failure, a donor heart, or heart and lungs, can be transplanted. Artificial hearts are occasionally used to keep patients alive whilst waiting for a heart transplant, or to allow the heart to rest as an aid to recovery.	WS 1.4 Evaluate given information about the advantages and disadvantages of treating cardiovascular diseases by drugs, mechanical devices or transplant. WS 1.3 Evaluate methods of treatment bearing in mind the benefits and risks associated with the treatment.

4.3.1.4 Homeostasis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the importance of maintaining a constant internal environment in response to internal and external change.	Homeostasis is the regulation of the internal conditions of a cell or organism to maintain optimum conditions for function in response to internal and external changes. Homeostasis is important because it maintains optimal conditions for enzyme action and all cell functions. Control of blood glucose concentration, control of body temperature and control of water levels in the human body are examples of homeostasis. An organism maintains homeostasis by monitoring its internal conditions and responding appropriately when these conditions deviate from their optimal state. These automatic control systems may involve nervous responses or chemical responses. Many of the processes are coordinated by hormones.	

4.3.1.5 Insulin and diabetes

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how insulin controls blood sugar levels in the body. (HT only) Explain how glucagon interacts with insulin to control blood sugar levels in the body.	Blood glucose concentration is monitored and controlled by the pancreas. If the blood glucose concentration is too high, the pancreas produces the hormone insulin, which causes glucose to move from the blood into the cells. In liver and muscle cells excess glucose is converted to glycogen for storage. (HT only) If the blood glucose concentration is too low, the pancreas produces glucagon, which causes glycogen to be converted into glucose and released into the blood.	MS 1a, 1c, 2c, 4a, 4c Translate information between numerical and graphical forms and extract and interpret information from graphs, charts and tables.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Compare Type 1 and Type 2 diabetes and explain how they can be treated.	Type 1 diabetes is a disorder in which the pancreas fails to produce sufficient insulin. It is characterised by uncontrolled high blood glucose levels and is normally treated with insulin injections. In Type 2 diabetes the body cells no longer respond to insulin produced by the pancreas. A carbohydrate controlled diet and an exercise regime are common treatments. Obesity is a risk factor for Type 2 diabetes.	

4.3.1.6 Human reproductive hormones

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the roles of hormones in human reproduction, including the menstrual cycle. (HT only) Explain the interactions of FSH, LH, oestrogen and progesterone in the control of the menstrual cycle.	During puberty reproductive hormones cause secondary sex characteristics to develop. Oestrogen is the main female reproductive hormone produced in the ovary. At puberty eggs begin to mature and one is released approximately every 28 days. This is called ovulation. Testosterone is the main male reproductive hormone produced by the testes and it stimulates sperm production. Several hormones are involved in the menstrual cycle of a woman. • Follicle-stimulating hormone (FSH) causes maturation of an egg in the ovary. • Luteinising hormone (LH) stimulates the release of the egg. • Oestrogen and progesterone are involved in maintaining the uterus lining.	MS 2c, 4a (HT only) Extract and interpret data from graphs showing hormone levels during the menstrual cycle.

4.3.1.7 Contraception

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the use of hormones in contraception and evaluate hormonal and non-hormonal methods of contraception.	Fertility can be controlled by a variety of hormonal and non-hormonal methods of contraception. These include: • oral contraceptives that contain hormones • injection, implant or skin patch of slow-release progesterone • barrier methods such as condoms and diaphragms • intrauterine devices • spermicidal agents • abstaining from intercourse at times when an egg may be fertilised • surgical methods of male and female sterilisation.	WS 1.4 Explain everyday and technological applications of science; evaluate associated personal, social, economic and environmental implications; and make decisions based on the evaluation of evidence and arguments.

4.3.1.8 Treatments for infertility (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the use of hormones in modern reproductive technologies to treat infertility.	The uses of hormones in controlling fertility include: • giving FSH and LH in a 'fertility drug' to a woman whose own level of FSH is too low • In Vitro Fertilisation (IVF) treatment, which involves giving a mother FSH and LH to stimulate the maturation of several eggs.	WS 1.4 Evaluate, from the perspective of patients and doctors, the methods of treating fertility bearing in mind that although fertility treatment gives couples the chance to have a baby of their own it is very emotionally and physically stressful; the success rates are not high and it can lead to multiple births which are a risk to both the babies and the mother.